

Disproof of Conjecture 000000002697

Pointwise Jacobian spectra miss tangent growth

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The logarithmic average of the pointwise Jacobian eigenvalues need not equal the Lyapunov exponent of the actual tangent evolution. A polynomial map of the affine plane provides a concrete counterexample; no independent cocycle certificate is assumed.

Define the actual map

$$F(x, y) = (1 + x - 6x^2 + 4x^3 + (1 - x)y, x^3 - x^2 + y).$$

It has a two-cycle $p_0 = (0, 0)$ and $p_1 = (1, 0)$, since $F(p_0) = p_1$ and $F(p_1) = p_0$. Its true derivative is

$$DF(x, y) = \begin{pmatrix} 1 - 12x + 12x^2 - y & 1 - x \\ 3x^2 - 2x & 1 \end{pmatrix}.$$

Consequently the Jacobians along the orbit alternate between

$$A = DF(p_0) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = DF(p_1) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Each complex spectrum is exactly $\{1\}$: in both cases $\det(zI - DF) = (z - 1)^2$. Thus every choice of pointwise Jacobian eigenvalues, and their logarithmic norm average along this orbit, gives zero.

However, $BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ has an expanding eigenvalue

$$\rho = \frac{3 + \sqrt{5}}{2} > 2, \quad \rho^2 - 3\rho + 1 = 0.$$

Let $v = (1, \rho - 1)$ and $w = Av = (\rho, \rho - 1)$. The actual tangent evolution $v_0 = v$, $v_{n+1} = DF(p_n)v_n$ satisfies

$$Bw = \rho v, \quad v_{2m} = \rho^m v, \quad v_{2m+1} = \rho^m w.$$

These identities follow from the actual derivative and orbit recurrence. The Zariski tangent spaces of the smooth affine plane are its usual two-dimensional tangent spaces, so this example also applies to the tangent-space wording of the source.

Use the maximum norm on the real plane. Then $\|v\| = \rho - 1$ and $\|w\| = \rho$, both positive. For every n ,

$$\log \|v_n\| = \left\lfloor \frac{n}{2} \right\rfloor \log \rho + r_n, \quad r_n \in \{\log(\rho - 1), \log \rho\}.$$

The remainder is bounded, while $\lfloor n/2 \rfloor / n \rightarrow 1/2$. Therefore the actual Lyapunov limit, along all iteration counts rather than just the even subsequence, is

$$\lim_{n \rightarrow \infty} \frac{\log \|v_n\|}{n} = \frac{\log \rho}{2} > 0.$$

This contradicts the pointwise spectral logarithmic average zero. In finite dimension changing the norm does not change this limit. Noncommuting products must be accounted for; pointwise eigenvalues alone do not determine tangent-vector growth. Here “Jacobi spectrum” is interpreted in its usual sense as the Jacobian eigenvalue spectrum. The separate symmetry-group assertion is not needed for the disproof.

Formal verification. Lean verifies the actual polynomial derivative, the two-cycle recurrence, the complete complex matrix spectrum at every orbit point, actual tangent iteration formulas, its logarithmic growth and the limit for all n , positivity of the resulting exponent, and zero logarithmic averages for every actual pointwise spectral selection. All nine final printed axiom audits use only `propext`, `Classical.choice` and `Quot.sound`.